CHAPTER 19 SUMMARY

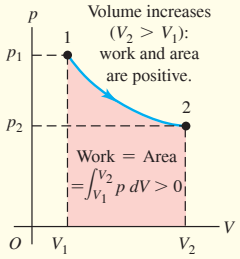
Heat and work in thermodynamic processes: A thermodynamic system has the potential to exchange energy with its surroundings by heat transfer or by mechanical work. When a system at pressure p changes volume from V_1 to V_2, it does an amount of work W given by the integral of p with respect to volume. If the pressure is constant, the work done is equal to p times the change in volume. A negative value of W means that work is done on the system. (See Example 19.1.)

In any thermodynamic process, the heat added to the system and the work done by the system depend not only on the initial and final states, but also on the path (the series of intermediate states through which the system passes).

$$W = \int_{V_1}^{V_2} p\,dV \qquad (19.2)$$

$$W = p(V_2 - V_1) \qquad (19.3)$$

(constant pressure only)



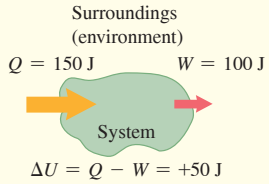
The first law of thermodynamics: The first law of thermodynamics states that when heat Q is added to a system while the system does work W, the internal energy U changes by an amount equal to $Q - W$. This law can also be expressed for an infinitesimal process. (See Examples 19.2, 19.3, and 19.5.)

The internal energy of any thermodynamic system depends only on its state. The change in internal energy in any process depends only on the initial and final states, not on the path. The internal energy of an isolated system is constant. (See Example 19.4.)

$$\Delta U = Q - W \qquad (19.4)$$

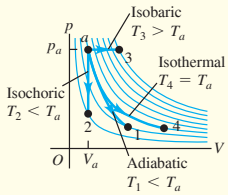
$$dU = dQ - dW \qquad (19.6)$$

(infinitesimal process)



Important kind of thermodynamic processes:

- Adiabatic process: No heat transfer into or out of a system; $Q = 0$.
- Isochoric process: Constant volume; $W = 0$.
- Isobaric process: Constant pressure; $W = p(V_2 - V_1)$.
- Isothermal process: Constant temperature.

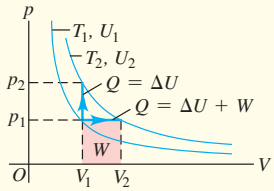


Thermodynamics of ideal gases: The internal energy of an ideal gas depends only on its temperature, not on its pressure or volume. For other substances the internal energy generally depends on both pressure and temperature.

The molar heat capacities C_V and C_p of an ideal gas differ by R, the ideal-gas constant. The dimensionless ratio of heat capacities, C_p/C_V, is denoted by γ. (See Example 19.6.)

$$C_p = C_V + R \qquad (19.17)$$

$$\gamma = \frac{C_p}{C_V} \qquad (19.18)$$

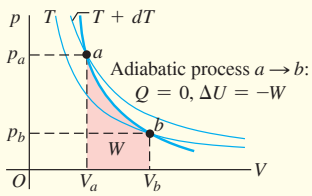


Adiabatic processes in ideal gases: For an adiabatic process for an ideal gas, the quantities $TV^{\gamma-1}$ and pV^{γ} are constant. The work done by an ideal gas during an adiabatic expansion can be expressed in terms of the initial and final values of temperature, or in terms of the initial and final values of pressure and volume. (See Example 19.7.)

$$W = nC_V(T_1 - T_2) \qquad (19.25)$$

$$= \frac{C_V}{R}(p_1V_1 - p_2V_2)$$

$$= \frac{1}{\gamma - 1}(p_1V_1 - p_2V_2) \qquad (19.26)$$



Chapter 19 Media Assets

For assigned homework and other learning materials, go to Mastering Physics.

GUIDED PRACTICE

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review EXAMPLES 19.2, 19.3, 19.4, and 19.5 (Section 19.4) before attempting these problems.

VP19.5.1 Consider the following thermodynamic processes. (a) The internal energy of a quantity of gas increases by 2.50×10^3 J as the gas does 3.25×10^3 J of work on its surroundings. How much heat flows into the gas? (b) Ice enclosed in a container does 2.50×10^4 J of work on its container as 7.00×10^3 J of heat flows out of the ice. What is the internal energy change of the ice? (c) The internal energy of a metal block increases by 4.25×10^3 J as 2.40×10^3 J of heat flows into the block. How much work did the block do on its surroundings?

VP19.5.2 How much work does a quantity of gas (volume V, pressure p) do in each of these processes? (a) V increases from 2.00×10^{-3} m^3 to 4.50×10^{-3} m^3 while p is a constant 6.20×10^4 Pa. (b) V is a constant 2.00×10^{-3} m^3 while p increases from 6.00×10^4 Pa to 9.00×10^4 Pa. (c) V decreases from 6.00×10^{-3} m^3 to 3.00×10^{-3} m^3 while p is a constant 1.25×10^5 Pa. (d) V increases from 2.00×10^{-3} m^3 to 4.50×10^{-3} m^3 while p increases from 1.50×10^5 Pa to 5.50×10^5 Pa. The graph of this process on a pV-diagram is a straight line.

VP19.5.3 A quantity of gas is enclosed in a cylinder that has a movable piston. The gas has volume $V = 3.60 \times 10^{-3}$ m^3 and pressure $p = 2.00 \times 10^5$ Pa. (a) First V decreases to 2.40×10^{-3} m^3 while p remains constant; then V remains constant while p increases to 6.00×10^5 Pa. During this process 1.56×10^3 J of heat flows into the gas. For this process, how much work does the gas do, and what is its internal energy change? (b) The gas returns to the same initial volume and pressure. Now p increases to 6.00×10^5 Pa while V remains constant; then V decreases to 2.40×10^{-3} m^3 while p remains constant. For this process, how much work does the gas do, and how much heat flows into the gas?

VP19.5.4 At an atmospheric pressure of 1.01×10^5 Pa, carbon dioxide (CO_2) sublimes at a temperature of -78.5°C. The heat of sublimation is 5.71×10^5 J/kg. You let heat flow into 6.0 kg of solid CO_2 (volume 4.0×10^{-3} m^3) at atmospheric pressure and a constant -78.5°C until it has completely sublimed to gas at -78.5°C. The CO_2 gas occupies a volume of 3.4 m^3. For this process, calculate (a) how much heat flows into the CO_2, (b) how much work the CO_2 does, and (c) the internal energy change of the CO_2.

Be sure to review EXAMPLE 19.6 (Section 19.7) before attempting these problems.

VP19.6.1 For neon gas, $C_V = \frac{3}{2}R$ and $C_p = \frac{5}{2}R$. Calculate the change in internal energy of 4.00 mol of neon if its temperature (a) increases from 20.0°C to 40.0°C at constant volume; (b) increases from 15.0°C to 30.0°C at constant pressure; (c) increases from 12.0°C to 24.0°C while both the pressure and volume increase.

VP19.6.2 For nitrogen gas, $C_V = \frac{5}{2}R$ and $C_p = \frac{7}{2}R$. You have 2.10 mol of nitrogen gas at temperature 27.0°C and pressure $p = 1.00 \times 10^5$ Pa. (a) Calculate the initial volume V of the gas. (b) Calculate the final temperature (in °C) of the gas and its internal energy change if (i) p doubles while V remains constant, (ii) V doubles while p remains constant, and (iii) both p and V double.

VP19.6.3 A quantity of argon gas ($C_V = \frac{3}{2}R$ and $C_p = \frac{5}{2}R$) is at pressure $p = 1.20 \times 10^5$ Pa and occupies volume $V = 0.250$ m^3. Calculate the change in internal energy of the gas if the final pressure and volume of the gas are (a) $p = 2.40 \times 10^5$ Pa, $V = 0.250$ m^3; (b) $p = 1.20 \times 10^5$ Pa, $V = 0.125$ m^3; (c) $p = 1.80 \times 10^5$ Pa, $V = 0.600$ m^3.

VP19.6.4 A quantity of n moles of oxygen gas ($C_V = \frac{5}{2}R$ and $C_p = \frac{7}{2}R$) is at absolute temperature T. You increase the absolute temperature to $2T$. Find the change in internal energy of the gas, the heat flow into the gas, and the work done by the gas if the process you used to increase the temperature is (a) isochoric, (b) isobaric, or (c) adiabatic.

Be sure to review EXAMPLE 19.7 (Section 19.8) before attempting these problems.

VP19.7.1 A quantity of argon gas ($C_V = \frac{3}{2}R$ and $C_p = \frac{5}{2}R$) is at pressure 4.00×10^5 Pa and occupies volume 2.00×10^{-3} m^3. Then the gas expands adiabatically to a new volume 6.00×10^{-3} m^3. Calculate (a) the value of γ for argon, (b) the final pressure of the gas, and (c) the work done by the gas as it expands.

VP19.7.2 A quantity of oxygen gas ($C_V = \frac{5}{2}R$ and $C_p = \frac{7}{2}R$) is at absolute temperature 325 K, is under pressure 1.00×10^5 Pa, and occupies volume 6.50×10^{-3} m^3. After you compress this gas adiabatically, its absolute temperature is 855 K. Calculate (a) the value of γ for oxygen, (b) the final volume of the gas, (c) the final pressure of the gas, and (d) the work done by the gas as it is compressed.

VP19.7.3 Initially 5.00 mol of neon gas ($C_V = \frac{3}{2}R$ and $\gamma = \frac{5}{2}R$) are at absolute temperature 305 K and occupy volume 4.00×10^{-2} m^3. Then the gas expands adiabatically to a new volume of 9.00×10^{-2} m^3. Calculate (a) the initial pressure of the gas, (b) the final pressure of the gas, (c) the final temperature of the gas, and (d) the work done by the gas as it expands.

VP19.7.4 You have 1.25 mol of hydrogen gas ($C_V = \frac{5}{2}R$ and $\gamma = \frac{7}{2}R$) at absolute temperature 325 K. You allow the gas to expand adiabatically to a final temperature of 195 K. (a) How much work does the gas do while being compressed? (b) What is the ratio of its final volume to its initial volume? (c) What is the ratio of the final gas pressure to the initial gas pressure?

BRIDGING PROBLEM Work Done by a van der Waals Gas

The van der Waals equation of state, an approximate representation of the behavior of gases at high pressure, is given by Eq. (18.7): $[p + (an^2/V^2)](V - nb) = nRT$, where a and b are constants having different values for different gases. (In the special case of $a = b = 0$, this is the ideal-gas equation.) (a) Calculate the work done by a gas with this equation of state in an isothermal expansion from V_1 to V_2. (b) For ethane gas (C_2H_6), $a = 0.554$ J·m^3/mol^2 and $b = 6.38 \times 10^{-5}$ m^3/mol. Calculate the work W done by 1.80 mol of ethane when it expands from 2.00×10^{-3} m^3 to 4.00×10^{-3} m^3 at a constant temperature of 300 K. Do the calculation by using (i) the van der Waals equation of state and (ii) the ideal-gas equation of state. (c) For which equation of state is W larger? Why should this be so?

SOLUTION GUIDE

IDENTIFY and SET UP

1. Review the discussion of the van der Waals equation of state in Section 18.1. What is the significance of the quantities a and b?

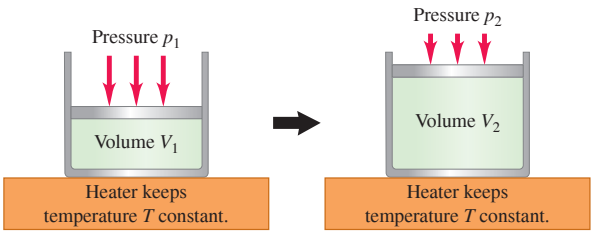
Continued

2. Decide how to find the work done by an expanding gas whose pressure p does not depend on V in the same way as for an ideal gas. (*Hint:* See Section 19.2.)
3. How will you find the work done by an expanding ideal gas?

EXECUTE

4. Find the general expression for the work done by a van der Waals gas as it expands from volume V_1 to volume V_2 (**Fig. 19.22**). (*Hint:* If you set $a = b = 0$ in your result, it should reduce to the expression for the work done by an expanding ideal gas.)
5. Use your result from step 4 to solve part (b) for ethane treated as a van der Waals gas.
6. Use the formula you chose in step 3 to solve part (b) for ethane treated as an ideal gas.

Figure **19.22** A gas undergoes an isothermal expansion.



EVALUATE

7. Is the difference between W for the two equations of state large enough to be significant?
8. Does the term with a in the van der Waals equation of state increase or decrease the amount of work done? What about the term with b? Which one is more important for the ethane in this problem?

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q19.1 For the following processes, is the work done by the system (defined as the expanding or contracting gas) on the environment positive or negative? (a) expansion of the burned petrol–air mixture in the cylinder of an automobile engine; (b) opening a bottle of champagne; (c) filling a scuba tank with compressed air; (d) partial crumpling of a sealed, empty water bottle as you drive from the mountains down to sea level.

Q19.2 It is not correct to say that an object contains a certain amount of heat, yet an object can transfer heat to another object. How can an object give away something it does not have in the first place?

Q19.3 In which situation must you do more work: inflating a balloon at sea level or inflating the same balloon to the same volume at the summit of Mt. Blanc? Explain in terms of pressure and volume change.

Q19.4 If you are told the initial and final states of a system and the associated change in internal energy, can you determine whether the internal energy change was due to work or to heat transfer? Explain.

Q19.5 Discuss the application of the first law of thermodynamics to a mountaineer who eats food, gets warm and perspires a lot during a climb, and does a lot of mechanical work in raising herself to the summit. The mountaineer also gets warm during the descent. Is the source of this energy the same as the source during the ascent?

Q19.6 When ice melts at 0°C, its volume decreases. Is the internal energy change greater than, less than, or equal to the heat added? How can you tell?

Q19.7 You hold an inflated balloon over a hot-air vent in your house and watch it slowly expand. You then remove it and let it cool back to room temperature. During the expansion, which was larger: the heat added to the balloon or the work done by the air inside it? Explain. (Assume that air is an ideal gas.) Once the balloon has returned to room temperature, how does the net heat gained or lost by the air inside it compare to the net work done on or by the surrounding air?

Q19.8 You bake chocolate chip cookies and put them, still warm, in a container with a loose (not airtight) lid. What kind of process does the air inside the container undergo as the cookies gradually cool to room temperature (isothermal, isochoric, adiabatic, isobaric, or some combination)? Explain.

Q19.9 Imagine a gas made up entirely of negatively charged electrons. Like charges repel, so the electrons exert repulsive forces on each other. Would you expect that the temperature of such a gas would rise, fall, or stay the same in a free expansion? Why?

Q19.10 In an adiabatic process for an ideal gas, the pressure decreases. In this process does the internal energy of the gas increase or decrease? Explain.

Q19.11 When you blow on the back of your hand with your mouth wide open, your breath feels warm. But if you partially close your mouth to form an "o" and then blow on your hand, your breath feels cool. Why?

Q19.12 An ideal gas expands while the pressure is kept constant. During this process, does heat flow into the gas or out of the gas? Justify your answer.

Q19.13 A liquid is irregularly stirred in a well-insulated container and thereby undergoes a rise in temperature. Regard the liquid as the system. Has heat been transferred? How can you tell? Has work been done? How can you tell? Why is it important that the stirring is irregular? What is the sign of ΔU? How can you tell?

Q19.14 When you use a hand pump to inflate the tires of your bicycle, the pump gets warm after a while. Why? What happens to the temperature of the air in the pump as you compress it? Why does this happen? When you raise the pump handle to draw outside air into the pump, what happens to the temperature of the air taken in? Again, why does this happen?

Q19.15 In the carburetor of an aircraft or automobile engine, air flows through a relatively small aperture and then expands. In cool, foggy weather, ice sometimes forms in this aperture even though the outside air temperature is above freezing. Why?

Q19.16 On a sunny day, large "bubbles" of air form on the sunwarmed earth, gradually expand, and finally break free to rise through the atmosphere. Soaring birds and glider pilots are fond of using these "thermals" to gain altitude easily. This expansion is essentially an adiabatic process. Why?

Q19.17 The prevailing winds on the Hawaiian island of Kauai blow from the northeast. The winds cool as they go up the slope of Mt. Waialeale (elevation 1523 m), causing water vapor to condense and rain to fall. There is much more precipitation at the summit than at the base of the mountain. In fact, with up to 350 rainy days per year, Mt. Waialeale is the rainiest spot on earth. But what makes the winds cool?

Q19.18 Applying the same considerations as in Question Q19.17, explain why the island of Niihau, a few kilometers to the southwest of Kauai, is almost a desert and farms there need to be irrigated.

Q19.19 In a constant-volume process, $dU = nC_V\,dT$. But in a constant-pressure process, it is *not* true that $dU = nC_p\,dT$. Why not?

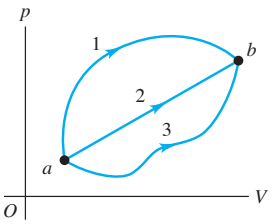
Q19.20 When a gas surrounded by air is compressed adiabatically, its temperature rises even though there is no heat input to the gas. Where does the energy come from to raise the temperature?

Q19.21 When a gas expands adiabatically, it does work on its surroundings. But if there is no heat input to the gas, where does the energy come from to do the work?

Q19.22 The gas used in separating the two uranium isotopes ^{235}U and ^{238}U has the formula UF_6. If you added heat at equal rates to a mole of UF_6 gas and a mole of H_2 gas, which one's temperature would you expect to rise faster? Explain.

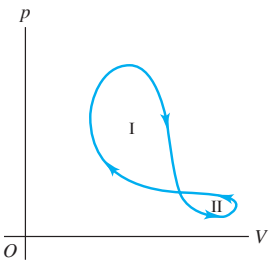
Q19.23 A system is taken from state a to state b along the three paths shown in **Fig. Q19.23**. (a) Along which path is the work done by the system the greatest? The least? (b) If $U_b > U_a$, along which path is the absolute value of the heat transfer, $|Q|$, the greatest? For this path, is heat absorbed or liberated by the system? Explain.

Figure **Q19.23**



Q19.24 A thermodynamic system undergoes a cyclic process as shown in **Fig. Q19.24**. The cycle consists of two closed loops: I and II. (a) Over one complete cycle, does the system do positive or negative work? (b) In each loop, is the net work done by the system positive or negative? (c) Over one complete cycle, does heat flow into or out of the system? (d) In each loop, does heat flow into or out of the system? Explain.

Figure **Q19.24**



EXERCISES

Section 19.2 Work Done During Volume Changes and Section 19.3 Paths Between Thermodynamic States

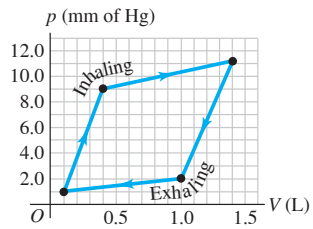
19.1 •• Two moles of an ideal gas are heated at constant pressure from $T = 27°C$ to $T = 107°C$. (a) Draw a pV-diagram for this process. (b) Calculate the work done by the gas.

19.2 • Six moles of an ideal gas are in a cylinder fitted at one end with a movable piston. The initial temperature of the gas is 26.2°C and the pressure is constant. As part of a machine design project, calculate the final temperature of the gas after it has done 1800 J of work.

19.3 •• **CALC** Two moles of an ideal gas are compressed in a cylinder at a constant temperature of 65.0°C until the original pressure has tripled. (a) Sketch a pV-diagram for this process. (b) Calculate the amount of work done.

19.4 •• **BIO** **Work Done by the Lungs.** The graph in **Fig. E19.4** shows a pV-diagram of the air in a human lung when a person is inhaling and then exhaling a deep breath. Such graphs, obtained in clinical practice, are normally somewhat curved, but we have modeled one as a set of straight lines of the same general shape. (*Important:* The pressure shown is the *gauge* pressure, *not* the absolute pressure.) (a) How many joules of *net* work does this person's lung do during one complete breath? (b) The process illustrated here is somewhat different from those we have been studying, because the pressure change is due to changes in the amount of gas in the lung, not to temperature changes. (Think of your own breathing. Your lungs do not expand because they've gotten hot.) If the temperature of the air in the lung remains a reasonable 20°C, what is the maximum number of moles in this person's lung during a breath?

Figure **E19.4**



19.5 •• **CALC** During the time 0.305 mol of an ideal gas undergoes an isothermal compression at 22.0°C, 516 J of work is done on it by the surroundings. (a) If the final pressure is 1.76 atm, what was the initial pressure? (b) Sketch a pV-diagram for the process.

19.6 •• A gas undergoes two processes. In the first, the volume remains constant at $0.200\ m^3$ and the pressure increases from 3.00×10^5 Pa to 4.50×10^5 Pa. The second process is a compression to a volume of $6.00 \times 10^{-2}\ m^3$ at a constant pressure of 4.50×10^5 Pa. (a) In a pV-diagram, show both processes. (b) Find the total work done by the gas during both processes.

19.7 • **Work Done in a Cyclic Process.** (a) In Fig. 19.7a, consider the closed loop $1 \rightarrow 3 \rightarrow 2 \rightarrow 4 \rightarrow 1$. This is a *cyclic* process in which the initial and final states are the same. Find the total work done by the system in this cyclic process, and show that it is equal to the area enclosed by the loop. (b) How is the work done for the process in part (a) related to the work done if the loop is traversed in the opposite direction, $1 \rightarrow 4 \rightarrow 2 \rightarrow 3 \rightarrow 1$? Explain.

Section 19.4 Internal Energy and the First Law of Thermodynamics

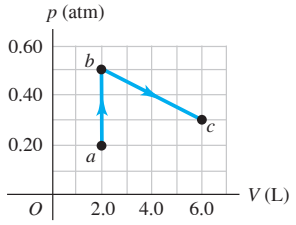
19.8 •• **CALC** An ideal gas undergoes a process during which the pressure is kept directly proportional to the volume, so that $p = \alpha V$, where α is a positive constant. If the volume changes from V_1 to V_2, how much work is done by the gas? Express your answer in terms of V_1, V_2, and α.

19.9 • A gas in a cylinder expands from a volume of $0.110\ m^3$ to $0.320\ m^3$. Heat flows into the gas just rapidly enough to keep the pressure constant at 1.65×10^5 Pa during the expansion. The total heat added is 1.15×10^5 J. (a) Find the work done by the gas. (b) Find the change in internal energy of the gas. (c) Does it matter whether the gas is ideal? Why or why not?

19.10 •• Five moles of an ideal monatomic gas with an initial temperature of 127°C expand and, in the process, absorb 1500 J of heat and do 2100 J of work. What is the final temperature of the gas?

19.11 •• The process *abc* shown in the *pV*-diagram in **Fig. E19.11** involves 0.0175 mol of an ideal gas. (a) What was the lowest temperature the gas reached in this process? Where did it occur? (b) How much work was done by or on the gas from *a* to *b*? From *b* to *c*? (c) If 215 J of heat was put into the gas during *abc*, how many of those joules went into internal energy?

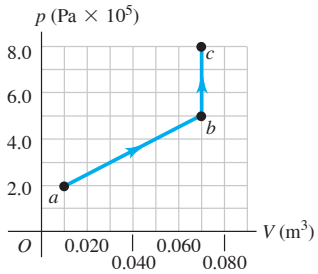
Figure **E19.11**



19.12 • A gas in a cylinder is held at a constant pressure of 2.28×10^5 Pa and is cooled and compressed from 1.69 m^3 to 1.18 m^3. The internal energy of the gas decreases by 1.50×10^5 J. (a) Find the work done by the gas. (b) Find the absolute value of the heat flow, $|Q|$, into or out of the gas, and state the direction of the heat flow. (c) Does it matter whether the gas is ideal? Why or why not?

19.13 •• The *pV*-diagram in **Fig. E19.13** shows a process *abc* involving 0.450 mol of an ideal gas. (a) What was the temperature of this gas at points *a*, *b*, and *c*? (b) How much work was done by or on the gas in this process? (c) How much heat had to be added during the process to increase the internal energy of the gas by 15,000 J?

Figure **E19.13**



19.14 • **Boiling Water at High Pressure.** When water is boiled at a pressure of 2.00 atm, the heat of vaporization is 2.20×10^6 J/kg and the boiling point is 120°C. At this pressure, 1.00 kg of water has a volume of 1.00×10^{-3} m^3, and 1.00 kg of steam has a volume of 0.824 m^3. (a) Compute the work done when 1.00 kg of steam is formed at this temperature. (b) Compute the increase in internal energy of the water.

19.15 • A thermodynamic system undergoes processes in which $|Q| = 100$ J and $|W| = 300$ J. Find Q and W, including whether the quantity is positive or negative, if the change in internal energy is (a) +400 J, (b) +200 J, (c) −200 J, and (d) −400 J.

Section 19.5 Kinds of Thermodynamic Processes, Section 19.6 Internal Energy of an Ideal Gas, and Section 19.7 Heat Capacities of an Ideal Gas

19.16 • During an isothermal compression of an ideal gas, 410 J of heat must be removed from the gas to maintain constant temperature. How much work is done by the gas during the process?

19.17 • A cylinder contains 0.250 mol of carbon dioxide (CO_2) gas at a temperature of 27.0°C. The cylinder is provided with a frictionless piston, which maintains a constant pressure of 1.00 atm on the gas. The gas is heated until its temperature increases to 127.0°C. Assume that the CO_2 may be treated as an ideal gas. (a) Draw a *pV*-diagram for this process. (b) How much work is done by the gas in this process? (c) On what is this work done? (d) What is the change in internal energy of the gas? (e) How much heat was supplied to the gas? (f) How much work would have been done if the pressure had been 0.50 atm?

19.18 • A cylinder contains 0.0100 mol of helium at $T = 27.0$°C. (a) How much heat is needed to raise the temperature to 67.0°C while keeping the volume constant? Draw a *pV*-diagram for this process. (b) If instead the pressure of the helium is kept constant, how much heat is needed to raise the temperature from 27.0°C to 67.0°C? Draw a *pV*-diagram for this process. (c) What accounts for the difference between your answers to parts (a) and (b)? In which case is more heat required? What becomes of the additional heat? (d) If the gas is ideal, what is the change in its internal energy in part (a)? In part (b)? How do the two answers compare? Why?

19.19 • In an experiment to simulate conditions inside an automobile engine, 0.185 mol of air at 780 K and 3.00×10^6 Pa is contained in a cylinder of volume 40.0 cm^3. Then 645 J of heat is transferred to the cylinder. (a) If the volume of the cylinder is constant while the heat is added, what is the final temperature of the air? Assume that the air is essentially nitrogen gas, and use the data in Table 19.1 even though the pressure is not low. Draw a *pV*-diagram for this process. (b) If instead the volume of the cylinder is allowed to increase while the pressure remains constant, repeat part (a).

19.20 •• A quantity of 5.00 mol of a monatomic ideal gas has an initial pressure of $p_a = 1.00 \times 10^5$ Pa and an initial volume of $V_a = 0.0200$ m^3.Two paths are used to take the gas to a final state for which $p_b = 1.80 \times 10^5$ Pa and $V_b = 0.0280$ m^3. (a) Along path 1 the pressure is increased to $p_a = 1.80 \times 10^5$ Pa while the volume is kept constant with a value of 0.0200 m^3. Then the volume is increased to 0.0280 m^3 while the pressure is kept at a constant 1.80×10^5 Pa. Calculate the total Q and total W for this path. (b) Along path 2 the volume is increased to 0.0280 m^3 while the pressure remains 1.00×10^5 Pa and then the pressure is increased to 1.80×10^5 Pa while the volume remains constant at 0.0280 m^3. Calculate the total Q and total W for this path. (c) Calculate the quantity $Q - W$ for each path. Your results should show that for these processes Q and W are path dependent but $Q - W$ is path independent.

19.21 • Heat Q flows into a monatomic ideal gas, and the volume increases while the pressure is kept constant. What fraction of the heat energy is used to do the expansion work of the gas?

19.22 • Three moles of an ideal monatomic gas expands at a constant pressure of 2.60 atm; the volume of the gas changes from 3.10×10^{-2} m^3 to 4.60×10^{-2} m^3. Calculate (a) the initial and final temperatures of the gas; (b) the amount of work the gas does in expanding; (c) the amount of heat added to the gas; (d) the change in internal energy of the gas.

19.23 • An experimenter adds 920 J of heat to 1.75 mol of an ideal gas to heat it from 10.0°C to 25.0°C at constant pressure. The gas does +240 J of work during the expansion. (a) Calculate the change in internal energy of the gas. (b) Calculate γ for the gas.

19.24 • A quantity of 2.00 mol of a monatomic ideal gas undergoes a compression during which the volume decreases from 0.0800 m^3 to 0.0500 m^3 while the pressure stays constant at a value of 1.80×10^4 Pa. (a) What is the work W? Is work done by the gas or on the gas? (b) What is the heat flow Q? Does heat enter or leave the gas? (c) What is the internal energy change for the gas? Does the internal energy of the gas increase or decrease?

19.25 • CALC The temperature of 0.140 mol of an ideal gas is held constant at 54.0°C while its volume is reduced to 21.0% of its initial volume. The initial pressure of the gas is 1.19 atm. (a) Determine the work done by the gas. (b) What is the change in its internal energy? (c) Does the gas exchange heat with its surroundings? If so, how much? Does the gas absorb or liberate heat?

19.26 •• For an ideal gas C_V and C_p are different because of the work W associated with the volume change for a constant-pressure process. To explore the difference between C_V and C_p for a liquid or a solid, consider the process in which 5.00 mol of ethanol is warmed from 10.0°C to 60.0°C while the applied pressure remains a constant 1.00 atm. The molar mass of ethanol is $M = 46.1$ g/mol. (a) What is the mass of 5.00 mol of ethanol? (b) How much heat Q enters the ethanol? (c) How much work W is done by the ethanol because of its thermal expansion? (d) What is the ratio W/Q?

Section 19.8 Adiabatic Processes for an Ideal Gas

19.27 • A monatomic ideal gas that is initially at 1.52×10^5 Pa and has a volume of 7.90×10^{-2} m^3 is compressed adiabatically to a volume of 4.50×10^{-2} m^3. (a) What is the final pressure? (b) How much work is done by the gas? (c) What is the ratio of the final temperature of the gas to its initial temperature? Is the gas heated or cooled by this compression?

19.28 •• Five moles of monatomic ideal gas have initial pressure 2.50×10^3 Pa and initial volume 2.10 m^3. While undergoing an adiabatic expansion, the gas does 1480 J of work. What is the final pressure of the gas after the expansion?

19.29 • During an adiabatic expansion the temperature of 0.500 mol of argon (Ar) drops from 52.0°C to 7.0°C. The argon may be treated as an ideal gas. (a) Draw a pV-diagram for this process. (b) How much work does the gas do? (c) What is the change in internal energy of the gas?

19.30 •• A player bounces a basketball on the floor, compressing it to 80.0% of its original volume. The air (assume it is essentially N_2 gas) inside the ball is originally at 20.0°C and 2.00 atm. The ball's inside diameter is 23.9 cm. (a) What temperature does the air in the ball reach at its maximum compression? Assume the compression is adiabatic and treat the gas as ideal. (b) By how much does the internal energy of the air change between the ball's original state and its maximum compression?

19.31 •• On a warm summer day, a large mass of air (atmospheric pressure 1.01×10^5 Pa) is heated by the ground to 30.0°C and then begins to rise through the cooler surrounding air. (This can be treated approximately as an adiabatic process; why?) Calculate the temperature of the air mass when it has risen to a level at which atmospheric pressure is only 9.00×10^4 Pa. Assume that air is an ideal gas, with $\gamma = 1.40$. (This rate of cooling for dry, rising air, corresponding to roughly 1°C per 100 m of altitude, is called the *dry adiabatic lapse rate.*)

19.32 • A cylinder contains 0.100 mol of an ideal monatomic gas. Initially the gas is at 1.00×10^5 Pa and occupies a volume of 2.50×10^{-3} m^3. (a) Find the initial temperature of the gas in kelvins. (b) If the gas is allowed to expand to twice the initial volume, find the final temperature (in kelvins) and pressure of the gas if the expansion is (i) isothermal; (ii) isobaric; (iii) adiabatic.

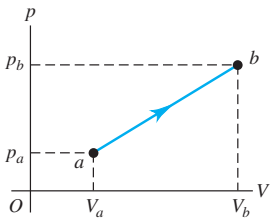
19.33 •• Helium gas undergoes an adiabatic process in which the Kelvin temperature doubles. By what factor does the pressure change?

19.34 • For a diatomic ideal gas, 3.01×10^{20} molecules undergo an adiabatic process in which the temperature increases from 25.0°C to 60.0°C. What is the change in the internal energy of the gas?

PROBLEMS

19.35 •• A quantity of air is taken from state a to state b along a path that is a straight line in the pV-diagram (**Fig. P19.35**). (a) In this process, does the temperature of the gas increase, decrease, or stay the same? Explain. (b) If $V_a = 6.60 \times 10^{-2}$ m^3, $V_b = 0.106$ m^3, $p_a = 1.05 \times 10^5$ Pa, and $p_b = 1.48 \times 10^5$ Pa, what is the work W done by the gas in this process? Assume that the gas may be treated as ideal.

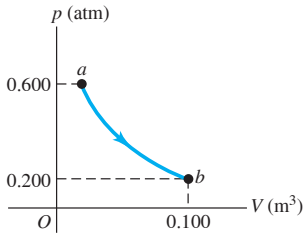
Figure P19.35



19.36 •• When water boils in a pot with a loose lid, the increased pressure forces the lid upward, releasing steam and reducing the pressure in the pot so that the lid drops back down. If the water continues to boil, the lid can rattle upward and downward. Consider a 1 L pot half filled with boiling water. In this case there is half a liter, or $\frac{1}{11.2}$ mol, of air and steam above the water. (a) Estimate the mass of the lid. (b) When the water is rapidly boiling, estimate the height to which the lid jumps during a steam-release event. (c) By multiplying the weight of the lid by the height of the jump, estimate the work done by the steam as the lid moves upward. (d) Treat the process as adiabatic and use the first law of thermodynamics to estimate the drop in the temperature of the steam as it pushes the lid upward.

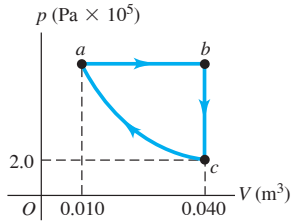
19.37 •• Figure P19.37 shows the pV-diagram for a process in which the temperature of the ideal gas remains constant at 85°C. (a) How many moles of gas are involved? (b) What volume does this gas occupy at a? (c) How much work was done by or on the gas from a to b? (d) By how much did the internal energy of the gas change during this process?

Figure P19.37



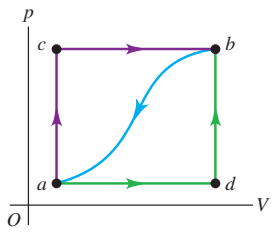
19.38 •• The graph in **Fig. P19.38** shows a pV-diagram for 3.25 mol of *ideal* helium (He) gas. Part ca of this process is isothermal. (a) Find the pressure of the He at point a. (b) Find the temperature of the He at points a, b, and c. (c) How much heat entered or left the He during segments ab, bc, and ca? In each segment, did the heat enter or leave? (d) By how much did the internal energy of the He change from a to b, from b to c, and from c to a? Indicate whether this energy increased or decreased.

Figure P19.38



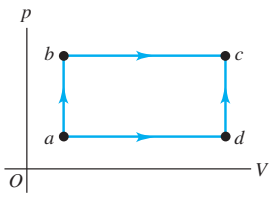
19.39 •• When a system is taken from state a to state b in **Fig. P19.39** along path acb, 85.0 J of heat flows into the system and 60.0 J of work is done by the system. (a) How much heat flows into the system along path adb if the work done by the system is 13.0 J? (b) When the system is returned from b to a along the curved path, the absolute value of the work done by the system is 39.0 J. Does the system absorb or liberate heat? How much heat? (c) If $U_a = 0$ and $U_d = 12.0$ J, find the heat absorbed in the processes ad and db.

Figure P19.39



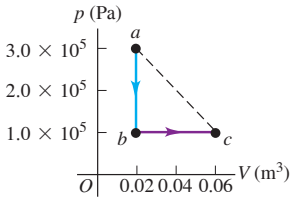
19.40 • A thermodynamic system is taken from state a to state c in **Fig. P19.40** along either path abc or path adc. Along path abc, the work W done by the system is 450 J. Along path adc, W is 120 J. The internal energies of each of the four states shown in the figure are $U_a = 150$ J, $U_b = 240$ J, $U_c = 680$ J, and $U_d = 330$ J. Calculate the heat flow Q for each of the four processes ab, bc, ad, and dc. In each process, does the system absorb or liberate heat?

Figure P19.40



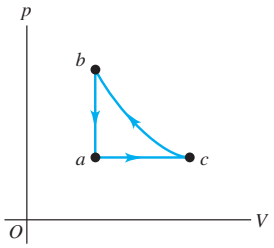
19.41 • A volume of air (assumed to be an ideal gas) is first cooled without changing its volume and then expanded without changing its pressure, as shown by path abc in **Fig. P19.41**. (a) How does the final temperature of the gas compare with its initial temperature? (b) How much heat does the air exchange with its surroundings during process abc? Does the air absorb heat or release heat during this process? Explain. (c) If the air instead expands from state a to state c by the straight-line path shown, how much heat does it exchange with its surroundings?

Figure P19.41



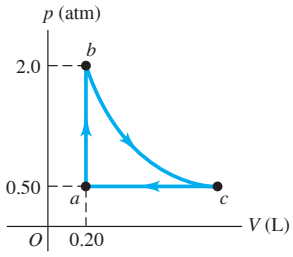
19.42 •• Three moles of an ideal gas are taken around cycle acb shown in **Fig. P19.42**. For this gas, $C_p = 29.1$ J/mol · K. Process ac is at constant pressure, process ba is at constant volume, and process cb is adiabatic. The temperatures of the gas in states a, c, and b are $T_a = 300$ K, $T_c = 492$ K, and $T_b = 600$ K. Calculate the total work W for the cycle.

Figure P19.42



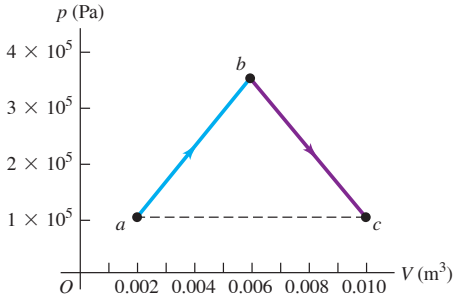
19.43 •• **Figure P19.43** shows a pV-diagram for 0.0040 mol of *ideal* H_2 gas. The temperature of the gas does not change during segment bc. (a) What volume does this gas occupy at point c? (b) Find the temperature of the gas at points a, b, and c. (c) How much heat went into or out of the gas during segments ab, ca, and bc? Indicate whether the heat has gone into or out of the gas. (d) Find the change in the internal energy of this hydrogen during segments ab, bc, and ca. Indicate whether the internal energy increased or decreased during each segment.

Figure P19.43



19.44 •• (a) One-third of a mole of He gas is taken along the path abc shown in **Fig. P19.44**. Assume that the gas may be treated as ideal. How much heat is transferred into or out of the gas? (b) If the gas instead went directly from state a to state c along the horizontal dashed line in Fig. P19.44, how much heat would be transferred into or out of the gas? (c) How does Q in part (b) compare with Q in part (a)? Explain.

Figure P19.44



19.45 •• **DATA** You place a quantity of gas into a metal cylinder that has a movable piston at one end. No gas leaks out of the cylinder as the piston moves. The external force applied to the piston can be varied to change the gas pressure as you move the piston to change the volume of the gas. A pressure gauge attached to the interior wall of the cylinder measures the gas pressure, and you can calculate the volume of the gas from a measurement of the piston's position in the cylinder.

You start with a pressure of 1.0 atm and a gas volume of 3.0 L. Holding the pressure constant, you increase the volume to 5.0 L. Then, keeping the volume constant at 5.0 L, you increase the pressure to 3.0 atm. Next you decrease the pressure linearly as a function of volume until the volume is 3.0 L and the pressure is 2.0 atm. Finally, you keep the volume constant at 3.0 L and decrease the pressure to 1.0 atm, returning the gas to its initial pressure and volume. The walls of the cylinder are good conductors of heat, and you provide the required heat sources and heat sinks so that the necessary heat flows can occur. At these relatively high pressures, you suspect that the ideal-gas equation will not apply with much accuracy. You don't know what gas is in the cylinder or whether it is monatomic, diatomic, or polyatomic. (a) Plot the cycle in the pV-plane. (b) What is the net heat flow for the gas during this cycle? Is there net heat flow into or out of the gas?

19.46 •• Nitrogen gas in an expandable container is cooled from 46.0°C to 11.3°C with the pressure held constant at 2.55×10^5 Pa. The total heat liberated by the gas is 2.55×10^4 J. Assume that the gas may be treated as ideal. Find (a) the number of moles of gas; (b) the change in internal energy of the gas; (c) the work done by the gas. (d) How much heat would be liberated by the gas for the same temperature change if the volume were constant?

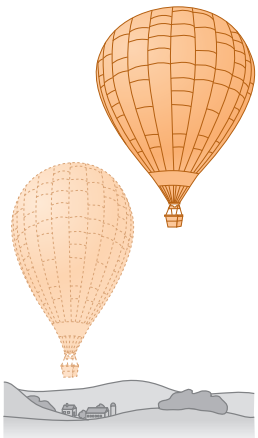
19.47 • **CALC** A cylinder with a frictionless, movable piston like that shown in Fig. 19.5 contains a quantity of helium gas. Initially the gas is at 1.00×10^5 Pa and 300 K and occupies a volume of 1.50 L. The gas then undergoes two processes. In the first, the gas is heated and the piston is allowed to move to keep the temperature at 300 K. This continues until the pressure reaches 2.50×10^4 Pa. In the second process, the gas is compressed at constant pressure until it returns to its original volume of 1.50 L. Assume that the gas may be treated as ideal. (a) In a pV-diagram, show both processes. (b) Find the volume of the gas at the end of the first process, and the pressure and temperature at the end of the second process. (c) Find the total work done by the gas during both processes. (d) What would you have to do to the gas to return it to its original pressure and temperature?

19.48 ••• **CP** A steel cargo drum has a height of 880 mm and a diameter of 610 mm. With its top removed it has a mass of 17.3 kg. The drum is turned upside down at the surface of the North Atlantic and is pulled downward into the ocean by a robotic submarine. On this day the surface temperature is 23.0°C and the surface air pressure is $p_0 = 101$ kPa. The water temperature decreases linearly with depth to 3.0°C at 1000 m below the surface. As the drum moves downward in the ocean, the air inside the drum is compressed, reducing the upward buoyant force. (a) At what depth $y_{neutral}$ is the barrel neutrally buoyant? (*Hint:* The pressure in the drum is equal to the sea pressure, which at depth y is $p_0 + \rho g y$, where $\rho = 1025\ \text{kg/m}^3$ is the density of seawater. The temperature at depth can be determined using the information above. Together with the ideal-gas law, you can derive a formula for the volume of air at depth y, and therefore a formula for the upward buoyant force as a function of depth.) (b) What is the volume of the air in the drum at depth $y_{neutral}$?

19.49 ••• **Chinook.** During certain seasons strong winds called chinooks blow from the west across the eastern slopes of the Rockies and downhill into Denver and nearby areas. Although the mountains are cool, the wind in Denver is very hot; within a few minutes after the chinook wind arrives, the temperature can climb 20°C ("chinook" refers to a Native American people of the Pacific Northwest). Similar winds occur in the Alps (called foehns) and in southern California (called Santa Anas). (a) Explain why the temperature of the chinook wind rises as it descends the slopes. Why is it important that the wind be fast moving? (b) Suppose a strong wind is blowing toward Denver (elevation 1630 m) from Grays Peak (80 km west of Denver, at an elevation of 4350 m), where the air pressure is 5.60×10^4 Pa and the air temperature is −11.0°C. The temperature and pressure in Denver before the wind arrives are 2.0°C and 8.12×10^4 Pa. By how many degrees Celsius will the temperature in Denver rise when the chinook arrives?

19.50 ••• **High-Altitude Research.** A large research balloon containing $2.00 \times 10^3\ \text{m}^3$ of helium gas at 1.00 atm and a temperature of 15.0°C rises rapidly from ground level to an altitude at which the atmospheric pressure is only 0.900 atm (**Fig. P19.50**). Assume the helium behaves like an ideal gas and the balloon's ascent is too rapid to permit much heat exchange with the surrounding air. (a) Calculate the volume of the gas at the higher altitude. (b) Calculate the temperature of the gas at the higher altitude. (c) What is the change in internal energy of the helium as the balloon rises to the higher altitude?

Figure **P19.50**



19.51 ••• An air pump has a cylinder 0.200 m long with a movable piston. The pump is used to compress air from the atmosphere (at absolute pressure 1.01×10^5 Pa) into a very large tank at 4.00×10^5 Pa gauge pressure. (For air, $C_V = 20.8\ \text{J/mol}\cdot\text{K}$.) (a) The piston begins the compression stroke at the open end of the cylinder. How far down the length of the cylinder has the piston moved when air first begins to flow from the cylinder into the tank? Assume that the compression is adiabatic. (b) If the air is taken into the pump at 28.0°C, what is the temperature of the compressed air? (c) How much work does the pump do in putting 20.0 mol of air into the tank?

19.52 •• A certain ideal gas has molar heat capacity at constant volume C_V. A sample of this gas initially occupies a volume V_0 at pressure p_0 and absolute temperature T_0. The gas expands isobarically to a volume $2V_0$ and then expands further adiabatically to a final volume $4V_0$. (a) Draw a pV-diagram for this sequence of processes. (b) Compute the total work done by the gas for this sequence of processes. (c) Find the final temperature of the gas. (d) Find the absolute value of the total heat flow $|Q|$ into or out of the gas for this sequence of processes, and state the direction of heat flow.

19.53 • A monatomic ideal gas expands slowly to twice its original volume, doing 400 J of work in the process. Find the heat added to the gas and the change in internal energy of the gas if the process is (a) isothermal; (b) adiabatic; (c) isobaric.

19.54 •• **CALC** A cylinder with a piston contains 0.250 mol of oxygen at 2.40×10^5 Pa and 355 K. The oxygen may be treated as an ideal gas. The gas first expands isobarically to twice its original volume. It is then compressed isothermally back to its original volume, and finally it is cooled isochorically to its original pressure. (a) Show the series of processes on a pV-diagram. Compute (b) the temperature during the isothermal compression; (c) the maximum pressure; (d) the total work done by the piston on the gas during the series of processes.

19.55 • Use the conditions and processes of Problem 19.54 to compute (a) the work done by the gas, the heat added to it, and its internal energy change during the initial expansion; (b) the work done, the heat added, and the internal energy change during the final cooling; (c) the internal energy change during the isothermal compression.

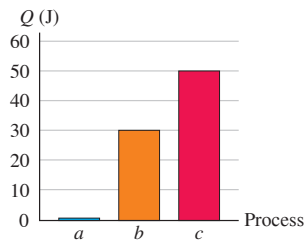
19.56 •• **CALC** A cylinder with a piston contains 0.152 mol of nitrogen at 1.78×10^5 Pa and 320 K. The nitrogen may be treated as an ideal gas. The gas is first compressed isobarically to half its original volume. It then expands adiabatically back to its original volume, and finally it is heated isochorically to its original pressure. (a) Show the series of processes in a pV-diagram. (b) Compute the temperatures at the beginning and end of the adiabatic expansion. (c) Compute the minimum pressure.

19.57 • Use the conditions and processes of Problem 19.56 to compute (a) the work done by the gas, the heat added to it, and its internal energy change during the initial compression; (b) the work done by the gas, the heat added to it, and its internal energy change during the adiabatic expansion; (c) the work done, the heat added, and the internal energy change during the final heating.

19.58 • **Comparing Thermodynamic Processes.** In a cylinder, 1.20 mol of an ideal monatomic gas, initially at 3.60×10^5 Pa and 300 K, expands until its volume triples. Compute the work done by the gas if the expansion is (a) isothermal; (b) adiabatic; (c) isobaric. (d) Show each process in a pV-diagram. In which case is the absolute value of the work done by the gas greatest? Least? (e) In which case is the absolute value of the heat transfer greatest? Least? (f) In which case is the absolute value of the change in internal energy of the gas greatest? Least?

19.59 •• **DATA** You have recorded measurements of the heat flow Q into 0.300 mol of a gas that starts at $T_1 = 20.0$°C and ends at a temperature T_2. You measured Q for three processes: one isobaric, one isochoric, and one adiabatic. In each case, T_2 was the same. **Figure P19.59** summarizes your results. But you lost a page from your lab notebook and don't have a record of the value of T_2; you also don't know which process was isobaric, isochoric, or adiabatic. Each process was done at a sufficiently low pressure for the gas to be treated as ideal. (a) Identify each process *a*, *b*, or *c* as isobaric, isochoric, or adiabatic. (b) What is the value of T_2? (c) How much work is done by the gas in each process? (d) For which process is the magnitude of the volume change the greatest? (e) For each process, does the volume of the gas increase, decrease, or stay the same?

Figure **P19.59**



19.60 •• **DATA** You compress a gas in an insulated cylinder—no heat flows into or out of the gas. The gas pressure is fairly low, so treating the gas as ideal is a good approximation. When you measure the pressure as a function of the volume of the gas, you obtain these results:

V (L)	2.50	2.02	1.48	1.01	0.50
p (atm)	0.101	0.139	0.202	0.361	0.952

(a) Graph $\log(p)$ versus $\log(V)$, with p in Pa and V in m^3. Explain why the data points fall close to a straight line. (b) Use your graph to calculate γ for the gas. Is the gas monatomic, diatomic, or polyatomic? (c) When $p = 0.101$ atm and $V = 2.50$ L, the temperature is 22.0°C. Apply the ideal-gas equation and calculate the temperature for each of the other pairs of p and V values. In this compression, does the temperature of the gas increase, decrease, or stay constant?

19.61 ••• A pneumatic shock absorber consists of a cylinder with a radius of 1.50 cm and a length that varies from a maximum of 30.0 cm to a minimum of 30.0 cm/v, where v is the compression ratio. For ease of manufacturing, v must be an integer. When the cylinder is fully extended and cooled, the air inside has an ambient pressure of 101 kPa and ambient temperature of 30.0°C. A quick adiabatic compression heats the air. The cylinder then expands isothermally until it reaches its maximum length. At that time the length remains fixed and the cylinder cools isochorically, returning to the ambient temperature. (a) How much work is done by the air during a full cycle, in terms of v? (b) If the air temperature cannot exceed 400°C, what is the maximum integer value of v? (c) If at least 25.0 J of work must be done by the gas in a full-compression cycle, what is the minimum allowable integer value of v? (d) What is the only permissible value of v? (e) How much heat leaves the air in the cylinder during the isochoric process if v has the value calculated in (d)?

CHALLENGE PROBLEM

19.62 •• **Engine Turbochargers and Intercoolers.** The power output of an automobile engine is directly proportional to the mass of air that can be forced into the volume of the engine's cylinders to react chemically with petrol. Many cars have a *turbocharger,* which compresses the air before it enters the engine, giving a greater mass of air per volume. This rapid, essentially adiabatic compression also heats the air. To compress it further, the air then passes through an *intercooler* in which the air exchanges heat with its surroundings at essentially constant pressure. The air is then drawn into the cylinders. In a typical installation, air is taken into the turbocharger at atmospheric pressure (1.01×10^5 Pa), density $\rho = 1.23$ kg/m^3, and temperature 15.0°C. It is compressed adiabatically to 1.45×10^5 Pa. In the intercooler, the air is cooled to the original temperature of 15.0°C at a constant pressure of 1.45×10^5 Pa. (a) Draw a pV-diagram for this sequence of processes. (b) If the volume of one of the engine's cylinders is 575 cm^3, what mass of air exiting from the intercooler will fill the cylinder at 1.45×10^5 Pa? Compared to the power output of an engine that takes in air at 1.01×10^5 Pa at 15.0°C, what percentage increase in power is obtained by using the turbocharger and intercooler? (c) If the intercooler is not used, what mass of air exiting from the turbocharger will fill the cylinder at 1.45×10^5 Pa? Compared to the power output of an engine that takes in air at 1.01×10^5 Pa at 15.0°C, what percentage increase in power is obtained by using the turbocharger alone?

MCAT-STYLE PASSAGE PROBLEMS

BIO Anesthetic Gases. One type of gas mixture used in anesthesiology is a 50%/50% mixture (by volume) of nitrous oxide (N_2O) and oxygen (O_2), which can be premixed and kept in a cylinder for later use. Because these two gases don't react chemically at or below 138 bar at typical room temperatures they form a homogeneous single gas phase, which can be considered an ideal gas. If the temperature drops below −6°C, however, N_2O may begin to condense out of the gas phase. Then any gas removed from the cylinder will initially be nearly pure O_2; as the cylinder empties, the proportion of O_2 will decrease until the gas coming from the cylinder is nearly pure N_2O.

19.63 In a test of the effects of low temperatures on the gas mixture, a cylinder filled at 20.0°C to 138 bar (gauge pressure) is cooled slowly and the pressure is monitored. What is the expected pressure at −5.00°C if the gas remains a homogeneous mixture? (a) 35 bar; (b) 105 bar; (c) 126 bar; (d) 132 bar.

19.64 In another test, the valve of a 500 L cylinder full of the gas mixture at 138 bar (gauge pressure) is opened wide so that the gas rushes out of the cylinder very rapidly. Why might some N_2O condense during this process? (a) This is an isochoric process in which the pressure decreases, so the temperature also decreases. (b) Because of the rapid expansion, heat is removed from the system, so the internal energy and temperature of the gas decrease. (c) This is an isobaric process, so as the volume increases, the temperature decreases proportionally. (d) With the rapid expansion, the expanding gas does work with no heat input, so the internal energy and temperature of the gas decrease.

19.65 You have a cylinder that contains 500 L of the gas mixture pressurized to 138 bar (gauge pressure). A regulator sets the gas flow to deliver 8.2 L/min at atmospheric pressure. Assume that this flow is slow enough that the expansion is isothermal and the gases remain mixed. How much time will it take to empty the cylinder? (a) 1 h; (b) 33 h; (c) 57 h; (d) 140 h.

19.66 In a hospital, pure oxygen may be delivered at 3.5 bar (gauge pressure) and then mixed with N_2O. What volume of oxygen at 20°C and 3.5 bar (gauge pressure) should be mixed with 1.7 kg of N_2O to get a 50%/50% mixture by volume at 20°C? (a) 0.21 m^3; (b) 0.27 m^3; (c) 1.9 m^3; (d) 100 m^3.

ANSWERS

Chapter Opening Question ?

(ii) The work done by a gas as its volume changes from V_1 to V_2 is equal to the integral $\int p\,dV$ between those two volume limits. If the volume of the gas contracts, the final volume V_2 is less than the initial volume V_1 and the gas does negative work. Propelling the locomotive requires that the gas do positive work, so the gas doesn't contribute to propulsion while contracting.

Key Example VARIATION Problems

VP19.5.1 (a) 5.75×10^3 J (b) -3.20×10^4 J (c) -1.85×10^3 J (d) 875 J

VP19.5.2 (a) 155 J (b) zero (c) -375 J (d) 875 J

VP19.5.3 (a) $W = -2.40 \times 10^2$ J, $\Delta U = 1.80 \times 10^3$ J (b) $W = -7.20 \times 10^2$ J, $Q = 1.08 \times 10^3$ J

VP19.5.4 (a) 3.4×10^6 J (b) 3.4×10^5 J (c) 3.1×10^6 J

VP19.6.1 (a) 998 J (b) 748 J (c) 599 J

VP19.6.2 (a) 5.24×10^{-2} m^3 (b) (i) 327°C, 1.31×10^4 J (ii) 327°C, 1.31×10^4 J (iii) 927°C, 3.93×10^4 J

VP19.6.3 (a) 4.50×10^4 J (b) -2.25×10^4 J (c) 1.17×10^5 J

VP19.6.4 (a) $\Delta U = \frac{5}{2}nRT$, $Q = \frac{5}{2}nRT$, $W = 0$ (b) $\Delta U = \frac{5}{2}nRT$, $Q = \frac{7}{2}nRT$, $W = nRT$ (c) $\Delta U = \frac{5}{2}nRT$, $Q = 0$, $W = -\frac{5}{2}nRT$

VP19.7.1 (a) $\frac{5}{3}$ (b) 6.41×10^4 Pa (c) 623 J

VP19.7.2 (a) $\frac{7}{5}$ (b) 5.79×10^{-4} m^3 (c) 2.95×10^6 Pa (d) -2.65×10^3 J

VP19.7.3 (a) 3.17×10^5 Pa (b) 8.20×10^4 Pa (c) 178 K (d) 7.94×10^3 J

VP19.7.4 (a) -3.38×10^3 J (b) 3.59 (c) 0.167

Bridging Problem

(a) $W = nRT \ln\left[\dfrac{V_2 - nb}{V_1 - nb}\right] + an^2\left[\dfrac{1}{V_2} - \dfrac{1}{V_1}\right]$

(b) (i) $W = 2.80 \times 10^3$ J, (ii) $W = 3.11 \times 10^3$ J

(c) ideal gas, for which there is no attraction between molecules